Signed Rational Numbers: Mixed Operations Review Answer Key

Grade 6–7 Mathematics

Quick Answers

1. -5	5. 2	9. $-\frac{3}{4}$
2. -7	6. $-\frac{9}{2}$	10. $>$
3. 28	7. $-\frac{7}{12}$	11. 1
4. $-\frac{3}{5}$	8. 3.75	12. $\frac{2}{5}$

Curriculum

Level: Grade 6–7 Mathematics
7.NS.A.1 – *problems 1, 2, 4, 5, 7, 8, 10*
7.NS.A.2 – *problems 3, 6, 9, 11, 12*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

5. *If they got* -16*: ignored the double negative and subtracted 9*
9. *If they got 0.75: dropped the negative sign on the product*

1. Evaluate: $-9+4$
Solution: The signs differ, so subtract the absolute values ($9-4=5$) and keep the sign of the number farther from zero, -9. $\boxed{-5}$

2. Evaluate: $4-11$
Solution: Rewrite as adding the opposite: $4-11=4+(-11)$. Different signs: $11-4=7$, and -11 is farther from zero, so the result is negative. $\boxed{-7}$

3. Evaluate: $(-4)(-7)$
Solution: A negative times a negative is positive: $4\cdot 7=28$. $\boxed{28}$

4. Evaluate: $-\frac{2}{5}+\left(-\frac{1}{5}\right)$
Solution: Same denominators and same signs: add the absolute values, $\frac{2}{5}+\frac{1}{5}=\frac{3}{5}$, and keep the negative sign. $\boxed{-\frac{3}{5}}$

5. Evaluate: $-7-(-9)$
Solution: Subtracting -9 is adding 9: $-7-(-9)=-7+9=2$. $\boxed{2}$

6. Evaluate, writing the answer as a fraction in simplest form: $-36 \div 8$

Solution: Different signs make the quotient negative: $\frac{36}{8} = \frac{9}{2}$ after dividing top and bottom by 4, so $-36 \div 8 = -\frac{9}{2}$. $\boxed{-\frac{9}{2}}$

7. Evaluate: $-\frac{5}{6} + \frac{1}{4}$

Solution: Use the common denominator 12: $-\frac{5}{6} = -\frac{10}{12}$ and $\frac{1}{4} = \frac{3}{12}$. Different signs: $\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$, negative because $-\frac{10}{12}$ is farther from zero. $\boxed{-\frac{7}{12}}$

8. Evaluate: $1.5 - (-2.25)$

Solution: Subtracting -2.25 is adding 2.25: $1.5 + 2.25 = 3.75$. $\boxed{3.75}$

9. Evaluate: $\left(-\frac{2}{3}\right)\left(\frac{9}{8}\right)$

Solution: Multiply numerators and denominators: $\frac{2 \cdot 9}{3 \cdot 8} = \frac{18}{24} = \frac{3}{4}$. One factor is negative, so the product is negative. $\boxed{-\frac{3}{4}}$

10. Write $<$ or $>$ in the blank: $-\frac{7}{8} \,\square\, -0.9$

Solution: Convert to the same form: $-\frac{7}{8} = -0.875$. On the number line -0.875 sits to the *right* of -0.9 (closer to zero), so $\boxed{-\frac{7}{8} \;>\; -0.9}$

11. Evaluate: $-2\left(\frac{3}{4} - \frac{5}{4}\right)$

Solution: Parentheses first: $\frac{3}{4} - \frac{5}{4} = -\frac{2}{4} = -\frac{1}{2}$. Then a negative times a negative: $-2 \cdot \left(-\frac{1}{2}\right) = 1$. $\boxed{1}$

12. **Challenge.** Evaluate: $\left(-\frac{3}{4} + \frac{1}{2}\right) \div \left(-\frac{5}{8}\right)$

Solution: Parentheses first: $-\frac{3}{4} + \frac{1}{2} = -\frac{3}{4} + \frac{2}{4} = -\frac{1}{4}$. Dividing by $-\frac{5}{8}$ means multiplying by its reciprocal $-\frac{8}{5}$:

$$-\frac{1}{4} \cdot \left(-\frac{8}{5}\right) = \frac{8}{20} = \frac{2}{5}.$$

Two negatives give a positive answer. $\boxed{\frac{2}{5}}$